

CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 8— Homework Assignment

This week's lecture previewed modern AI — multimodal models, LLMs, AI agents, fine-tuning, LoRA, and RAG — ahead of your midterm. Before you go further into that material, this assignment asks you to lock down a skill you will be tested on directly next week: **calculating Support, Confidence, Lift, Leverage, and Conviction by hand**. The scenario below is brand new — a home-goods store, not the one used in class or in any lecture example — so working through it will genuinely test whether you can apply the formulas yourself, not just recall a worked answer.

1. Scenario: HemHome — A Home-Goods Store

Code	Item
A	Table Lamp
B	Pillow Cover
C	Ceramic Bowl
D	Dinner Plate
E	Curtain Panel
F	Scented Candle

The store recorded the following $N = 8$ transactions during a Saturday afternoon:

Transaction	Items Purchased
T1	Table Lamp (A), Pillow Cover (B), Ceramic Bowl (C), Curtain Panel (E)
T2	Pillow Cover (B), Ceramic Bowl (C), Dinner Plate (D)
T3	Table Lamp (A), Curtain Panel (E), Scented Candle (F)
T4	Pillow Cover (B), Curtain Panel (E)
T5	Table Lamp (A), Dinner Plate (D), Scented Candle (F)
T6	Ceramic Bowl (C), Dinner Plate (D)
T7	Table Lamp (A), Curtain Panel (E)
T8	Ceramic Bowl (C), Scented Candle (F)

$N = 8$ transactions.

2. Your Task

All questions concern the rule **{Pillow Cover} $\Rightarrow$ {Ceramic Bowl}** — i.e. $\{B\} \Rightarrow \{C\}$. Show every step of your arithmetic; a final number with no working will receive partial credit at most.

2.1 Support, Confidence, and Lift

1. $\text{Support}(B)$ — the fraction of transactions containing Pillow Cover
2. $\text{Support}(C)$ — the fraction of transactions containing Ceramic Bowl
3. $\text{Support}(B, C)$ — the fraction of transactions containing both items together
4. $\text{Confidence}(\{B\} \Rightarrow \{C\})$
5. $\text{Lift}(\{B\} \Rightarrow \{C\})$

2.2 Leverage and Conviction

Using the same rule $\{B\} \Rightarrow \{C\}$, calculate Leverage and Conviction. Show the formula you used before you substitute numbers.

2.3 Is Confidence Symmetric?

Now calculate Confidence in the opposite direction: $\{C\} \Rightarrow \{B\}$ (Ceramic Bowl $\Rightarrow$ Pillow Cover). Is this value equal to the Confidence you found in Question 2.1? In 2–3 sentences, explain why Confidence does or does not have the symmetric property, using your own two numbers as evidence.

2.4 Interpreting Lift

Based on the Lift value you calculated in 2.1, do you think the relationship between Pillow Cover and Ceramic Bowl is meaningful, or could it plausibly have arisen by chance? Justify your answer using what Lift actually measures.

2.5 A Business Recommendation

Propose one concrete action HemHome's store manager could take based on this rule — for example, product placement, a bundle/combo price, or a promotional display. One or two sentences is enough, but be specific to these two products.



A Note on Grading

Full marks require correct, fully-shown arithmetic for every sub-question in 2.1 and 2.2 — not just a correct final answer. Questions 2.3–2.5 are graded on the quality of your reasoning, not on matching a single "correct" sentence. Round decimals to 3 significant figures where needed, and keep fractions or decimals consistent within an answer.

3. Submission Instructions

- Work through all of Section 2 by hand, on paper — this mirrors exactly how you'll need to work during the midterm, with no computer or calculator app.
- Write your Student ID and full name clearly at the top of your first page.
- Take a clear photo or scan of every page (make sure numbers and units are legible).
- Combine your pages into a single PDF (or a single clear JPG per page if a PDF isn't available).
- Upload your file to the Week 8 Homework submission point in the course system before the deadline posted there.

*Good luck — and treat this as a rehearsal for the exam,
not just an assignment.*

P'Kao